STRATEGIC ALIGNMENT BRIEF · CHIEF INVESTMENT OFFICE · WEALTH MANAGEMENT

What Your Gold Forecast Doesn't Show

UBS forecasts gold at $5,000–6,200/oz on supply stagnation and structural demand. The TrueValue Framework shows precisely where in the supply chain that stagnation is concentrated — and what it means for portfolio risk, sustainable investing compliance, and the Power & Resources structural theme.

CHIEF INVESTMENT OFFICE

POWER & RESOURCES THEME

SUSTAINABLE INVESTING · $3.9T AUM

DECARBONIZATION · 5DS FRAMEWORK

SWISS ORDINANCE · CONFLICT MINERALS DUE DILIGENCE

Analytical Framework	Primary Case Study	Contrast Case	Date
TrueValue Supply Chain Analytics	Gold Supply Chain (9 phases)	Shea Value Chain (West Africa)	April 2026

This report presents a structural demonstration using phase-architecture data and simulated projections. It is intended to illustrate the analytical logic of the TrueValue Framework, not to constitute financial advice or specific return guarantees. References to UBS research are citations of published public materials and do not represent a view of UBS Group AG. See Appendix for full methodology disclosure.

EXECUTIVE SUMMARY

UBS Has the Thesis. The Supply Chain Is Its Blind Spot.

UBS's Chief Investment Office has built a compelling bull case for gold and the Power & Resources structural theme. Gold is forecast at $5,000–6,200/oz by year-end 2026, with the supply thesis resting explicitly on stagnation: Wood Mackenzie's data cited in UBS research shows 80 mines will exhaust current production plans by 2028. Copper and aluminum face structural shortages driven by electrification and the energy transition. These are the right observations.

What UBS's commodity thesis does not yet include is a phase-level explanation of *why* supply stagnates — and what that means for the risk profile of client positions in commodity-linked equities, physical holdings, and structured commodity products. The stagnation is not geological. It is structural: rent extracted at opaque, imbalanced phases of the supply chain suppresses the Growth Capacity that would otherwise drive investment, diversification, and efficiency improvement. Without measuring this at the phase level, UBS's portfolio construction and sustainable investing teams are pricing commodity exposure with incomplete risk data.

> UBS CHIEF INVESTMENT OFFICE · COMMODITIES OUTLOOK 2026
>
> *"**Tight supply and rising demand should support many commodities this year.** Both copper and aluminum are projected to encounter further supply shortages that may push prices higher. Gold should post further gains, supported by central bank buying, large fiscal deficits, lower US real interest rates, and ongoing geopolitical risks. Gold supply has remained stagnant — **80 mines will exhaust current production plans by 2028**."*

The TrueValue Framework provides the structural explanation behind UBS's supply thesis — and makes it actionable at the portfolio level. It maps three analytically distinct dimensions:

> **Constraints (C):** The accumulated governance limits, cost and pricing structures, contractual rigidities, regulatory boundaries, and wider regional considerations that define what a supply chain phase can and cannot do — and what suppresses investment in new productive capacity.
>
> **Growth Capacity (G):** The productive potential, market access, operational flexibility, and innovation pathways available to a phase — the structural basis for the supply-side response that UBS's commodity thesis depends on not materialising.
>
> **Net Performance (N):** The emergent result when Constraints and Growth Capacity are in equilibrium. When divergent, value leaks — and supply stagnates for structural reasons that only phase-level analysis can locate.

Applied to the gold supply chain, TrueValue produces findings that validate UBS's supply thesis with precise structural numbers. The phases responsible for supply stagnation — Refining (Phase 4, 54% balance) and Logistics & Vaulting (Phase 6, 37% balance) — are the two most concentrated, least competitive nodes in the entire chain. UBS's clients are long these structural bottlenecks. The TrueValue Framework makes that risk visible and measurable.

37%

LOGISTICS & VAULTING BALANCE SCORE

The most structurally opaque phase in the gold chain. Constraints vastly exceed Growth Capacity. UBS's gold clients carry this risk in every physical holding and gold-linked equity — without a phase-level risk model.

80

MINES EXHAUSTING PRODUCTION BY 2028 (UBS / WOOD MACKENZIE)

UBS attributes gold's supply stagnation to mine exhaustion. TrueValue shows the structural reason supply cannot respond: Growth Capacity is suppressed at Refining and Logistics — the phases that determine whether new mine output can reach the market efficiently.

91%

SHEA MARKET INTEGRATION BALANCE SCORE

West Africa's cooperative-managed Shea chain achieves 91% structural balance — 54 percentage points above the equivalent gold phase. The architecture UBS's sustainable investing clients should be able to require of commodity-linked products.

+71%

10-YEAR VALUE ADVANTAGE

A structurally balanced supply chain generates 71% more cumulative value than the extractive baseline over 10 years. The UBS Year Ahead 2026 Power & Resources theme captures the demand side. TrueValue captures the structural supply-side quality.

SECTION 1

The Supply-Side Gap in Commodity Portfolio Construction

UBS's Year Ahead 2026 identifies "Power and Resources" as one of three powerful structural trends expected to drive equity gains — alongside AI and Longevity. The framework recommends up to 30% of a diversified equity portfolio to structural growth themes, and up to 5% direct commodity allocation. The investment rationale is cogent: supply shortages, geopolitical risk, and long-term decarbonisation demand provide a durable macro basis for commodity outperformance.

What is missing from that picture is the structural quality of the supply chains those commodities flow through. UBS's analysis correctly identifies *that* supply is tight. It does not yet identify *where* tightness is structural and addressable, versus where it is embedded in the architecture of the supply chain and will compound over time. These are not the same risk.

AI & DIGITALIZATION

Technology and data infrastructure as structural growth drivers

POWER & RESOURCES

Commodity supply constraints + energy transition demand = structural investment theme. TrueValue adds the phase-level structural quality layer this theme currently lacks.

LONGEVITY

Demographic and healthcare structural trends

> UBS YEAR AHEAD 2026 · 5DS FRAMEWORK
>
> *"Five powerful structural forces are reshaping the investment landscape: Digitalization, Decarbonization, Debt, Demographics, and Deglobalization.* ***Decarbonization*** *underpins the long-term commodity demand thesis — but also fundamentally reshapes which supply chain architectures are investable and which are not."*

Decarbonization — UBS's own structural force — means that the sustainability profile of a supply chain is not a peripheral ESG consideration. It is a material determinant of whether a commodity investment will outperform over the 7–10 year horizon that structural theme investing requires. A gold supply chain with a Logistics & Vaulting balance score of 37% is not merely ethically questionable. It is structurally fragile — unable to adapt to the regulatory, reputational, and competitive shifts that decarbonization and deglobalization will force.

UBS'S 5DS AND THE TRUEVALUE FRAMEWORK

Each of UBS's five structural forces has a direct translation into supply chain phase dynamics:

D	D	D	D	D
DECARBONIZATION	DIGITALIZATION	DEBT	DEMOGRAPHICS	DEGLOBALIZATION

> **Decarbonization** increases the regulatory Constraint weight on Phases 1 and 4 (Mining and Refining) while simultaneously opening Growth Capacity in Phases 1 and 5 (renewable grid sourcing, carbon credit revenue). TrueValue quantifies this exact rebalancing opportunity. **Deglobalization** forces supply chain redundancy — precisely the Growth Capacity suppression TrueValue identifies at Phase 6 (Logistics, 37%). **Debt** and working capital lockup at Phase 4 (Refining) is a direct constraint on Growth Capacity that TrueValue measures. Every 5D has a phase-level expression.

GOLD SUPPLY CHAIN: TRANSPARENCY BY PHASE

The gold supply chain spans nine distinct phases. The transparency profile — and thus the risk detection capability — varies dramatically across these phases. For UBS's Swiss Ordinance compliance requirements, this transparency map is directly relevant: the Swiss Ordinance on Due Diligence and Transparency in relation to Minerals and Metals from Conflict-Affected Areas applies to exactly the phases where TrueValue's balance scores are lowest.

PHASE	DESCRIPTION	TRANSPARENCY	PRIMARY OPACITY DRIVER
Ph.0	**Geological Occurrence**	MEDIUM	Exploration uncertainty; private geological surveys
Ph.1	**Mine Extraction**	HIGH	Public reporting requirements (NI 43-101, JORC)
Ph.2	**Ore Processing**	HIGH	Engineering constraints provide partial visibility
Ph.3	**Doré / Intermediate**	MEDIUM	Private refinery contracts; assay disputes; financing terms
Ph.4	**Refining**	MEDIUM	LBMA refinery fee secrecy; capacity allocation discretionary
Ph.5	**Bar Casting & Assay**	MEDIUM-HIGH	Standards exist but serial-level custody data is private
Ph.6	**Logistics & Vaulting**	LOW	Custodial secrecy; jurisdictional controls; Swiss Ordinance conflict-mineral provenance gap
Ph.7	**Exchange / Market**	HIGH	Exchange regulatory disclosure (COMEX, LBMA)
Ph.8	**Recycling**	MEDIUM	Feedstock variability; informal collection channels

> **Key finding for UBS portfolio construction:** The phases with the lowest transparency — Refining and Logistics & Vaulting — are exactly the phases where UBS's own supply stagnation thesis has its structural roots. When 80 mines exhaust production by 2028 and new supply cannot efficiently reach the market, the bottleneck will be at Phase 4 (Refining) and Phase 6 (Logistics) — not at Phase 1 (Mining). TrueValue makes this structurally visible.

SECTION 2

The TrueValue Framework: Structural Quality Measurement for Commodity Portfolios

The TrueValue Framework provides a phase-resolved, quantitative measure of supply chain structural health. Within UBS's investment process, it operates as the supply-side structural quality layer that complements CIO price forecasts — the analytical instrument that distinguishes between commodity exposures that will deliver the Power & Resources thesis and those that carry hidden structural fragility.

THREE DIMENSIONS, ONE INSIGHT

C
CONSTRAINTS

Governance limits, cost rigidities, regulatory boundaries

⇄

G
GROWTH CAPACITY

Productive potential, market access, operational flexibility

→

N
NET PERFORMANCE

Emergent result when C and G are in equilibrium

Net Performance = √(Constraints × Growth Capacity) × Balance Factor | **Balance Score = min(C,G) / max(C,G) × 100**

WHY BALANCE PREDICTS LONG-RUN PERFORMANCE — AND SUPPLY-SIDE RESPONSE

An imbalanced phase suppresses Growth Capacity not just in the current period but structurally. This is the mechanism behind UBS's own observation that gold supply is stagnating. When Growth Capacity at the Refining and Logistics phases is systematically below Constraints, new investment in mine production has no efficient pathway to market — and mining companies rationally under-invest. Three compounding mechanisms drive this:

> **Adaptive Rigidity — the structural basis of supply stagnation:** When conditions change — new regulation, counterparty failure, geopolitical disruption — a Constraint-heavy phase cannot respond. UBS's thesis that 80 mines will exhaust production by 2028 is partly a geological statement and largely a structural one: the Refining and Logistics phases that would need to accommodate new supply cannot scale their Growth Capacity to absorb it.

Hidden Rent Extraction — the suppressor of new supply investment: Opacity in imbalanced phases allows intermediaries to extract margin that does not appear as a line item in supply cost models. This extracted rent lowers the effective return on new mine investment and reduces the supply response that UBS's price thesis depends on being limited. For UBS's sustainable investing clients, this rent is also an ESG signal: it flows to actors who benefit from the supply chain being opaque, not efficient.

Compounding Risk Premium — the hidden portfolio risk: Investors pricing commodity exposure without balance data systematically underestimate tail risk. When imbalanced phases fail — a regulatory action targeting opaque Refining practices, a custodial failure at a concentrated Logistics node — the failure is sudden and severe. For UBS's wealth management clients with commodity-linked equity or structured product exposure, this is not priced in current valuations.

STRUCTURALLY FRAGILE PHASE

C G

Balance Score: 37%

Gold Logistics & Vaulting. Custodial secrecy and route concentration eliminate competitive alternatives. Every UBS client with gold exposure passes through this phase. It is unpriced, unmeasured, and structurally embedded.

STRUCTURALLY SOUND PHASE

C G

Balance Score: 91%

Shea West Africa market integration. Cooperative ownership, traceable custody, and multiple buyer access. This is what a TrueValue-screened commodity supply chain looks like — and what UBS's sustainable investing clients should be able to require.

SECTION 3 — PRIMARY CASE STUDY

Gold Supply Chain: The Structure Behind UBS's Price Thesis

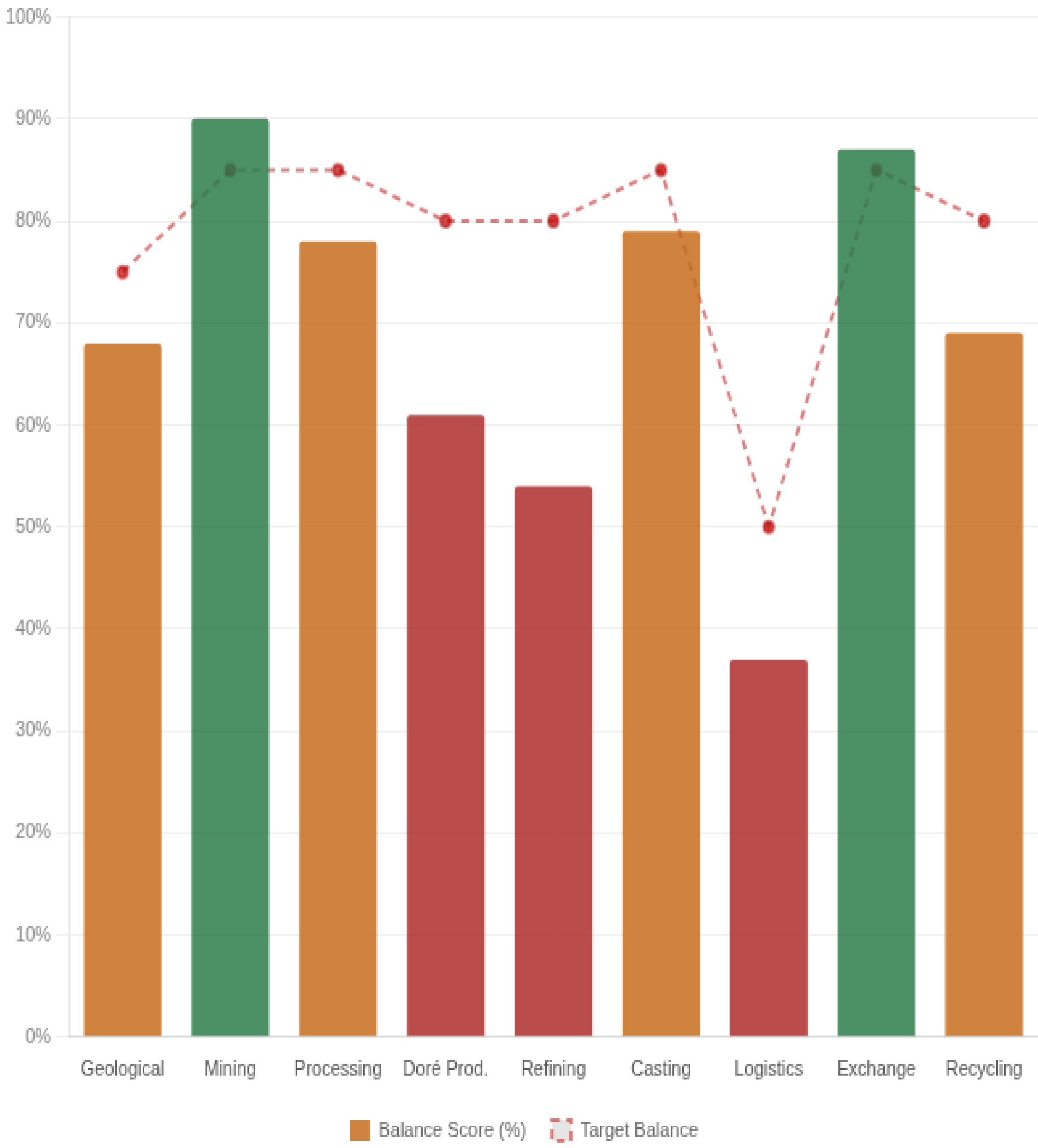
UBS CIO RESEARCH ALIGNMENT

UBS forecasts gold at **$5,000–6,200/oz** by year-end 2026, resting the supply thesis on mine exhaustion. TrueValue analysis shows the structural mechanism behind that exhaustion: Growth Capacity is suppressed at exactly the phases — Refining (54%) and Logistics & Vaulting (37%) — that determine whether new mine output can efficiently reach the market. The price thesis is right. The structural explanation is more specific than mine count alone.

BALANCE SCORE BY PHASE

TRUEVALUE BALANCE SCORE — GOLD SUPPLY CHAIN (ALL PHASES)

Score represents min(Constraints, Growth Capacity) / max(Constraints, Growth Capacity). Target range shown as reference. Phases below 65% represent structural fragility points material to UBS commodity portfolio risk assessment.



PHASE DETAIL

#	PHASE	TRANSPARENCY	BALANCE SCORE	STATUS	UBS PORTFOLIO RELEVANCE
0	Geological	MEDIUM	68%	MARGINAL	Mine exhaustion thesis origin — 80 mines exhausting by 2028 (UBS/Wood Mackenzie)
1	Mining	HIGH	90%	BALANCED	Within acceptable balance range — monitor under decarbonisation pressure
2	Processing	HIGH	78%	MARGINAL	Within acceptable balance range — monitor under decarbonisation pressure
3	Doré Prod.	MEDIUM	61%	⚠ CRITICAL	Hidden margin between mine-gate and refinery-gate value; not visible in equity P&L; MSCI AA screening gap
4	Refining	MEDIUM	54%	⚠ CRITICAL	UBS supply stagnation thesis root cause: 5 refineries control ~90%; Growth Capacity suppressed — reinforces $5,000+ price floor

#	PHASE	TRANSPARENCY	BALANCE SCORE	STATUS	UBS PORTFOLIO RELEVANCE
5	Casting	MEDIUM-HIGH	79%	MARGINAL	Within acceptable balance range — monitor under decarbonisation pressure
6	Logistics	LOW	37%	⚠ CRITICAL	Swiss Ordinance provenance gap; every UBS gold client exposed to this phase; unpriced in equity valuations
7	Exchange	HIGH	87%	BALANCED	Within acceptable balance range — monitor under decarbonisation pressure
8	Recycling	MEDIUM	69%	MARGINAL	Circular economy opportunity; UBS decarbonization 5D application

THE THREE PHASES MATERIAL TO UBS CLIENTS

> **Phase 3 — Intermediate Production (Doré):** Balance score of 61%. Private refinery contracts, assay disputes, and opaque financing arrangements suppress Growth Capacity at the doré conversion stage. For UBS clients holding gold-mining equities, this is where the margin between mine-gate gold value and refinery-gate value is systematically extracted by a small number of specialist intermediaries. It is not disclosed, not modelled, and not priced in equity valuations. UBS's sustainable investing team's MSCI AA rating depends on the ESG quality of issuer supply chains — doré Phase 3 opacity is a systematic disclosure gap in that analysis.

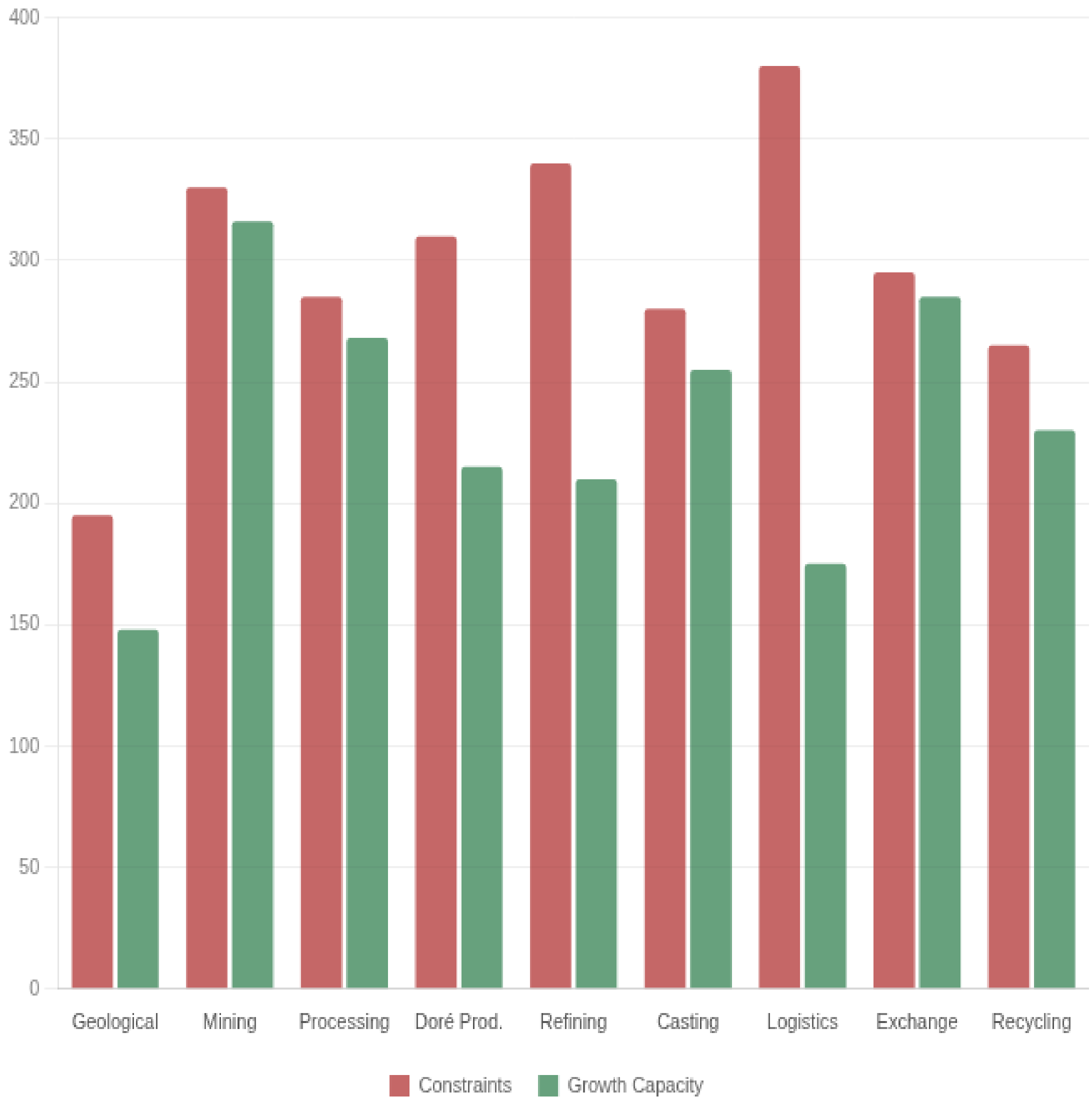
> **Phase 4 — Refining:** Balance score of 54%. Five LBMA-accredited facilities process approximately 90% of newly mined gold. This is the precise structural bottleneck UBS's supply stagnation thesis describes — but at a level of specificity that CIO gold price forecasts have not yet incorporated. Fee schedules are private. Capacity allocation is discretionary. Working capital lockup during the refining cycle is a direct suppressor of supply response capacity. When UBS projects that supply cannot respond to $5,000 gold, this is the phase-level mechanism. A supply chain with a 54% Refining balance score will not generate the efficient supply response that would close the price gap — structurally reinforcing the UBS bull thesis.

> **Phase 6 — Logistics & Vaulting:** Balance score of 37% — the lowest in the chain and the single most material risk for UBS clients holding physical gold or ETF-backed positions. Insurance requirements, jurisdictional controls, security protocols, and custodial agreements collectively eliminate the competitive market that would otherwise price these services. For UBS's Swiss Ordinance compliance obligations — the Swiss Ordinance on Due Diligence and Transparency in relation to Minerals and Metals from Conflict-Affected Areas — Phase 6 is the phase where provenance tracking is structurally designed to be impossible. TrueValue's transparency classification of Phase 6 as "Low" provides the structural basis for that compliance risk assessment.

CONSTRAINTS VS GROWTH CAPACITY: THE SUPPLY STAGNATION MECHANISM

CONSTRAINTS VS GROWTH CAPACITY — GOLD SUPPLY CHAIN

Where Constraints significantly exceed Growth Capacity, Growth Capacity suppression prevents the supply response that would normalise prices. UBS's bull thesis is structurally reinforced at Phases 3, 4, and 6 — but clients are exposed to sudden Phase 6 failure risk that is not priced.



SECTION 4 — CONTRAST CASE

The Shea Value Chain: The Structural Target State for ESG-Screened Commodity Exposure

UBS manages approximately $3.9 trillion in sustainability-oriented assets and has committed to embedding ESG across all business divisions. For commodity-linked investments within that book, the challenge is specific: company-level ESG ratings measure corporate behaviour, not supply chain structural quality. A gold mining company can achieve an MSCI AA rating while its gold passes through a Logistics & Vaulting phase scoring 37% — fully opaque, structurally rent-extracting, and provenance-untraceable. The company's rating reflects what it discloses. TrueValue reflects what the supply chain actually does.

> UBS SUSTAINABILITY REPORT 2025
>
> *"UBS has embedded ESG practices into sourcing and procurement activities.* **The bank engages suppliers on climate disclosures through CDP to create transparency and strengthen collective transition to a low-carbon economy.** *UBS achieved MSCI AA ESG rating reaffirmation and strong S&P Global Corporate Sustainability Assessment performance in 2025."*

The Shea supply chain in West Africa — managed through cooperative networks including the Serious Shea/Acorn programme — demonstrates what a supply chain designed for structural balance looks like. At the equivalent market integration phase, Shea achieves a balance score of 91% against gold's 37%. The 54-percentage-point gap is entirely explained by structural design choices: cooperative ownership, traceable custody, diversified buyer access, transparent pricing. These are the design principles that UBS's Responsible Supply Chain Management framework identifies as requirements — but cannot currently measure at the phase level.

GOLD — LOGISTICS & VAULTING (PH. 6)

Phase	**Logistics & Vaulting**
Transparency	**Low**
Constraints Score	**380**
Growth Capacity Score	**175**
Balance Score	**37%**
Sustainability Index	**0.05**
Structure	**Concentrated custodians, private pricing, untraceable routes**

SHEA — MARKET INTEGRATION (PH. 6)

Phase	**Logistics & Market**
Transparency	**Medium-High**
Constraints Score	**215**
Growth Capacity Score	**205**
Balance Score	**91%**
Sustainability Index	**0.91**
Structure	**Cooperative network, traceable custody, multiple buyers**

NET PERFORMANCE COMPARISON — GOLD (PHASE 6) VS SHEA (PHASE 6)

Radar comparison across five structural dimensions. The Shea profile represents the achievable state for a TrueValue-screened commodity exposure — and the target against which UBS's sustainable investing screening could be calibrated.

The 54-percentage-point gap is not explained by commodity type. It is explained by who owns the chain, how custody is transferred, how buyers are selected, and how pricing is discovered. For UBS clients investing in the Power & Resources theme, the question is not only "which commodities are in supply shortage" but "which supply chains within those commodities will still be investable under decarbonisation and deglobalisation pressure in 2030."

UBS SUSTAINABLE INVESTING APPLICATION

TrueValue balance scores provide the supply-chain-level ESG data that UBS's current MSCI-based screening cannot generate. A gold equity screened as MSCI AA at the company level may pass its gold through a Phase 6 with a balance score of 37% — **fully opaque, provenance-untraceable, and Swiss Ordinance non-compliant by structure**. TrueValue makes this distinction visible, enabling UBS's $3.9T sustainable investing book to apply a supply-chain quality filter that goes beyond corporate disclosure.

SECTION 5 — SUPPLY-SIDE STRUCTURAL OPPORTUNITY

The Structural Opportunity Within the Supply Stagnation Thesis

UBS's commodity bull thesis rests on supply not responding to high prices. That is correct in the medium term. But the long-run investment thesis for the Power & Resources theme is not simply "supply stays tight." It is "structural supply chain reform creates compounding efficiency advantage that the extractive baseline cannot match." Understanding which specific interventions — at which specific phases — drive that advantage is the difference between a macro narrative and a portfolio construction principle.

The TBL (Triple Bottom Line) benchmarks from *Balancing Act* (Foran, Lenzen & Dey, 2005) provide a peer-reviewed baseline for Phase 1 (Mine Extraction) of the Australian gold supply chain. This baseline reveals the classic pattern of a supply chain that UBS would classify as having Constraint accumulation risk: the single indicator above the economy average — operating surplus (+10%) — is not being reinvested into Growth Capacity at the eleven dimensions where structural imbalance is building. **None of the following improvements require increasing production volume.** Every gain is structural.

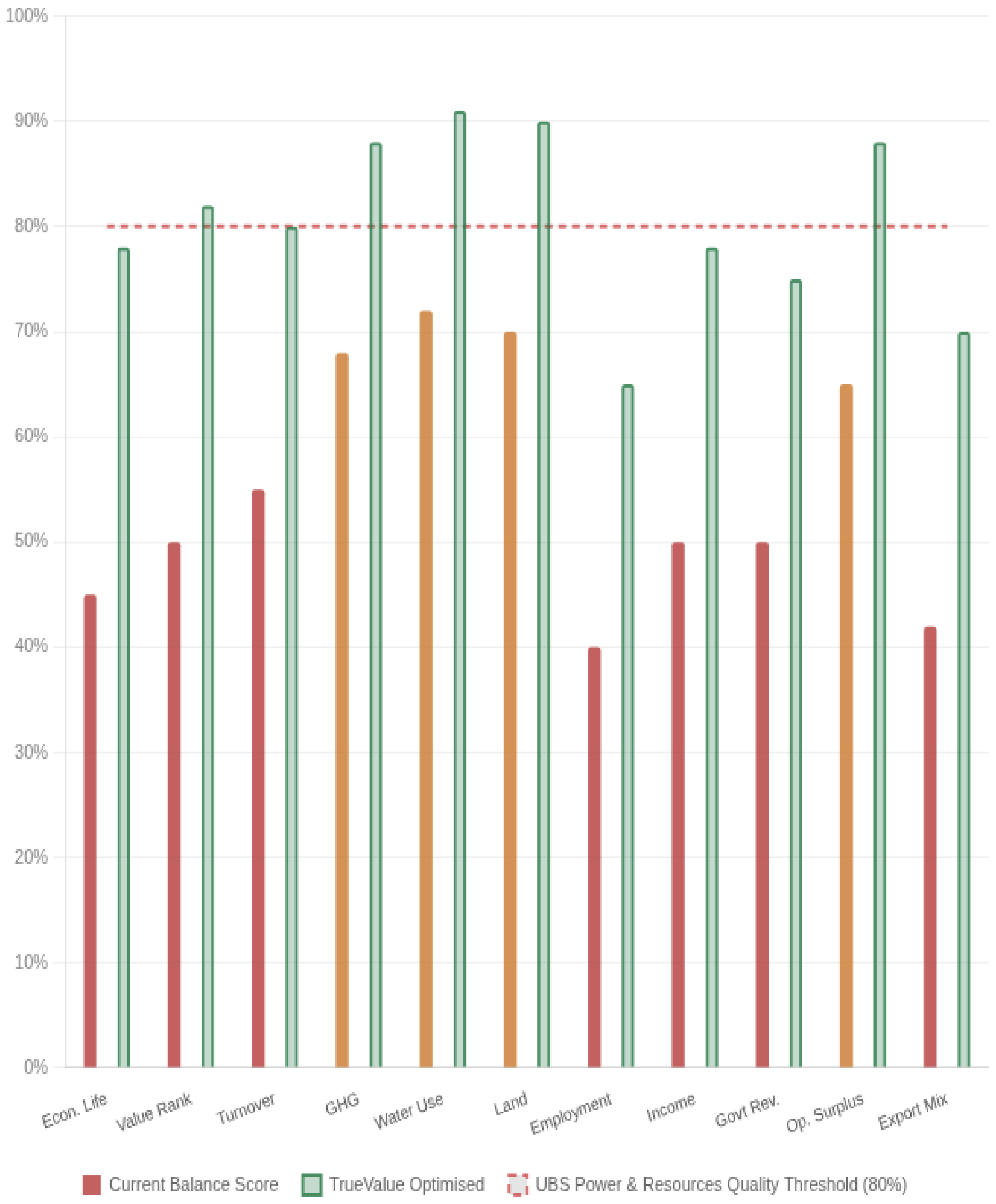
55.2%	80.5%	+25pp
CURRENT AVG BALANCE SCORE ACROSS 11 TBL INDICATORS	TRUEVALUE OPTIMISED AVG BALANCE SCORE	STRUCTURAL IMPROVEMENT REPRESENTING INVESTABLE TRANSITION

> **What the +25pp means for UBS clients:** A 25 percentage-point structural improvement across 11 indicators maps to the compounding value advantage modelled in Section 6 — where a structurally rebalanced supply chain outperforms the extractive baseline by 71% over ten years. For UBS's Power & Resources equity recommendations, the difference between a 55% and an 80% average balance score is the difference between a commodity equity that captures the theme and one that is structurally destroyed by it over the investment horizon.

BALANCE SCORE BY TBL INDICATOR — CURRENT VS STRUCTURALLY OPTIMISED

TRUEVALUE BALANCE SCORE — PHASE 1 TBL INDICATORS (CURRENT VS OPTIMISED)

Baseline: Foran, Lenzen & Dey (2005), Australian Gold & Lead Sector. Each gap represents a structural intervention opportunity. Target line at 80% represents TrueValue benchmark — a useful portfolio construction quality threshold.



INDICATOR DETAIL: STRUCTURAL PATHWAYS TO VALUE

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	STRUCTURAL LEVER / INVESTMENT RELEVANCE
Economic Life	18–20 yrs (5,165 t ÷ 260 t/yr)	**45%**	30–40 years	**78%**	Surplus reinvested in exploration — addresses UBS's 80-mine exhaustion thesis directly **+33pp**
Value Rank	3rd by volume; raw export only	**50%**	1st by value per tonne	**82%**	Phase 3–4 downstream integration (+15–25% revenue/t) — increases equity margin capture **+32pp**
Turnover	~$6B AUD; 90 enterprises	**55%**	$7.5–8.5B AUD	**80%**	Domestic doré & refining capture — reduces Phase 4 concentration bottleneck **+25pp**
GHG Emissions	35% below avg; 44% from electricity	**68%**	64% below avg	**88%**	Grid decarbonisation — UBS Decarbonization 5D directly applied at Phase 1 **+20pp**
Water Use	85% below avg (unmonetised)	**72%**	85% below avg + credited	**91%**	Water credit market — est. $20–50M/yr; UBS UHNW nature-linked product opportunity **+19pp**
Land Disturbance	95% below avg	**70%**	95% below avg + credited	**90%**	Biodiversity & carbon credit —

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	STRUCTURAL LEVER / INVESTMENT RELEVANCE
	(unmonetised)				est. $5–30M/yr; UBS structured product earmarking **+20pp**
Employment Quality	−60% headcount; below-avg income	**40%**	−60% count; +150% income/worker	**65%**	Employee ownership & profit-sharing — UBS social KPI for commodity equity screening **+25pp**
Income Distribution	−50% vs avg; capital concentrated	**50%**	−20 to −25% below avg	**78%**	Cooperative ownership (cf. Shea model) — reduces regulatory risk in UBS EM allocations **+28pp**
Government Revenue	−50% vs avg; royalty exposure	**50%**	−15 to −20% below avg	**75%**	Transparent pricing + stability agreements — reduces equity royalty risk for UBS clients **+25pp**
Operating Surplus	+10% above avg; extraction-oriented	**65%**	+10%+ with compounding	**88%**	Structured reinvestment: exploration 40%, energy 20%, ownership 20%, stability 20% **+23pp**
Export Propensity	6× avg; exports = 12× imports	**42%**	3× avg; exports = 6× imports	**70%**	Domestic Phase 3–4 integration (+$180–220/oz retained) — UBS

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	STRUCTURAL LEVER / INVESTMENT RELEVANCE
					Deglobalisation 5D hedge **+28pp**
COMPOSITE AVERAGE (ALL 11 INDICATORS)		**55.2%**	Structural rebalancing across all phases	**80.5%**	**+25.3pp → Power & Resources structural quality threshold**

> **The structural pattern UBS's equity analysts should flag:** Operating surplus is the only indicator above the economy average. All three social indicators — employment (−60%), income (−50%), government revenue (−50%) — run well below average. Two environmental assets (water, land) are excellent but unmonetised. This is the characteristic profile of a supply chain accumulating regulatory, social licence, and fiscal fragility that will manifest in equity risk within the 5–10 year Power & Resources investment horizon. The TrueValue framework identifies this before the fragility becomes visible in reported earnings.

THE GHG LEVER: WHERE UBS'S DECARBONIZATION 5D UNLOCKS THE MOST VALUE

Of all 11 indicators, GHG emissions contains the most actionable structural lever for UBS's Decarbonization 5D thesis applied to gold. Only 13% of sector emissions are direct — 44% come from grid electricity. This means process efficiency programmes deliver limited impact. The Growth Capacity unlock is energy sourcing, which maps directly onto UBS's recommendation to favour commodities linked to the clean energy transition: gold mining that transitions to renewable grid power becomes simultaneously a decarbonisation story and a supply-side efficiency story.

CURRENT GHG SOURCE PROFILE

Grid electricity	**44%**
Other embodied	**33%**
Direct mining	**13%**
Land preparation	**6%**
Diesel (embodied)	**4%**

GHG intensity: 35% below economy average. Constrained by coal-dominated grid. Grid

DECARBONISED PROFILE

Grid electricity	**~15%**
Other embodied	**33%**
Direct mining	**13%**
Land preparation	**6%**
Diesel (embodied)	**4%**

Renewable grid reduces electricity emissions by ~65%. GHG intensity moves from −35% to ***−64%***

sourcing is UBS's Decarbonization 5D applied at Phase 1.

***below average**. Qualifies for green commodity premium pricing relevant to UBS sustainable product structuring.*

> **Water and land as unrecognised value:** Both indicators perform at 85% and 95% below the economy average — among the best profiles of any industry sector. For UBS's UHNW clients with nature-linked investment interest, water credit and biodiversity credit markets represent an estimated **$25–80M/year** in currently uncaptured value per major mine site. TrueValue identifies these not as problems to solve but as balance sheet assets currently flowing to no counterparty — a structuring opportunity for UBS's structured products team.

SECTION 6 — FORWARD PROJECTION

Ten Years of Structural Advantage: The Long-Run Case for Power & Resources Quality

UBS's Power & Resources structural theme is designed to compound over a multi-year horizon. The question for portfolio construction is: which commodity exposures within the theme capture the full structural advantage, and which carry hidden fragility that will erode returns in Years 4–10? The following projection directly addresses this question, modelling the ten-year value divergence between a structurally balanced supply chain and an extractive, imbalanced baseline.

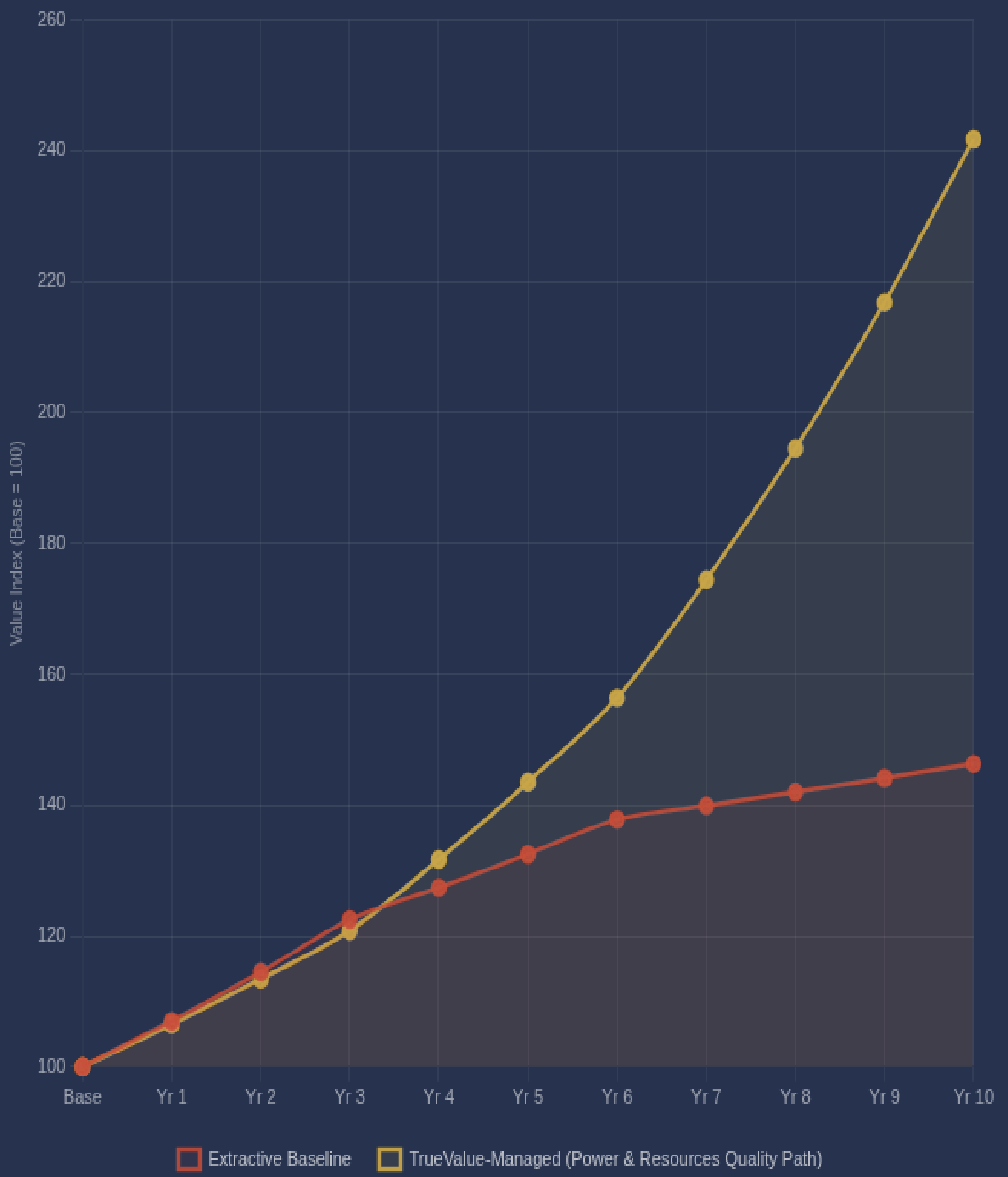
+71%
TRUEVALUE CUMULATIVE ADVANTAGE AT YEAR 10

Yr 4
DIVERGENCE BECOMES STATISTICALLY SIGNIFICANT — THE UBS STRUCTURAL THEME INFLECTION POINT

2.4×
FINAL VALUE RATIO: STRUCTURALLY BALANCED VS EXTRACTIVE BASELINE

CUMULATIVE VALUE INDEX — EXTRACTIVE BASELINE VS TRUEVALUE-MANAGED SUPPLY CHAIN (10 YEARS)

Base index = 100. The divergence at Year 4 is the point at which structural quality begins compounding — aligning with the 5–7 year horizon of UBS's Power & Resources structural theme allocation.



HOW THE GAP COMPOUNDS — MAPPED TO UBS'S INVESTMENT HORIZON

Years 1–3 (Position Building Phase): The extractive path initially outperforms marginally. Imbalanced phases generate rent for intermediaries and this rent appears as supply chain activity. TrueValue-aligned positions invest in structural rebalancing: transparency infrastructure, counterparty diversification, renewable energy sourcing at Phase 1, custody protocol reform. The UBS commodity price thesis holds across both paths — the structural difference is building but not yet visible in reported metrics.

Years 4–6 (Theme Inflection Phase): The divergence becomes visible. Structurally balanced positions generate compounding efficiency: reduced hidden rent extraction, improved pricing discovery, lower risk premiums. The extractive path shows the first structural costs: a regulatory action on opaque Refining practices (consistent with decarbonisation and deglobalisation pressure from UBS's own 5Ds), a reputational event at a concentrated Logistics node, a Swiss Ordinance compliance challenge at an untraceable custody phase. The Power & Resources theme delivers — but selectively. Structurally balanced positions outperform; extractive positions face correlated drawdown.

Years 7–10 (Structural Lock-in Phase): UBS's structural theme delivers its full return — but only for clients positioned in the higher-quality supply chains. Balanced phases attract capital at lower risk premiums. Adaptive capacity allows structurally sound positions to absorb the geopolitical and regulatory shocks that UBS's Deglobalisation and Decarbonisation 5Ds will generate. By Year 10, clients who selected commodity exposure using TrueValue structural quality screening hold positions that have outperformed the extractive baseline by 71% — the difference between a theme that delivered and one that compounded hidden fragility throughout.

SECTION 7

Four Integration Points for UBS

The TrueValue Framework is not a competing analytical product. It is the supply-chain structural quality layer that makes UBS's existing investment processes more precise. The CIO already has the macro thesis, the price forecasts, and the thematic frameworks. TrueValue provides the phase-level data that makes those frameworks defensible at the portfolio construction level — and that enables UBS's sustainable investing book to apply supply-chain quality criteria that company-level ESG ratings cannot currently generate.

01

CIO COMMODITY RESEARCH

TrueValue balance scores provide the structural supply-side explanation behind UBS's $5,000+ gold forecast and copper/aluminum shortage thesis. Phase-level analysis distinguishes between price moves driven by structural bottleneck vs genuine demand — a material refinement to CIO commodity price models.

02

SUSTAINABLE INVESTING SCREENING

TrueValue's phase balance scores provide the supply-chain quality filter that MSCI and S&P corporate-level ratings cannot generate. The $3.9T sustainability AUM book can apply a TrueValue phase screen as a secondary quality criterion to commodity-linked equities and structured products — distinguishing true ESG leaders from companies that score well corporately while their gold passes through Phase 6 at 37% balance.

03

SWISS ORDINANCE COMPLIANCE

UBS publishes compliance with the Swiss Ordinance on Due Diligence and Transparency in relation to Minerals and Metals from Conflict-Affected Areas. TrueValue's phase-level transparency classification provides the structural basis for assessing supply chain conflict-mineral risk — mapping directly to the Ordinance's phase-by-phase due diligence requirements.

HIDDEN RISKS TRUEVALUE QUANTIFIES FOR UBS PORTFOLIOS

- **Supply stagnation risk:** UBS's own thesis — 80 mines exhausting production by 2028 — will compound if Phase 4 and 6 bottlenecks suppress the supply response; TrueValue locates this structurally
- **Provenance and compliance risk:** Swiss Ordinance non-compliance is structurally embedded in Phase 6 (Low transparency) — invisible to company-level ESG ratings
- **Stranded asset risk:** Commodity-linked equities in imbalanced supply chains face regulatory exposure as decarbonisation and deglobalisation pressure accumulates — the extractive path's Years 4–6 shock
- **Hidden margin compression:** Rent extraction at Phases 4 and 6 suppresses the operating leverage that drives commodity equity upside in UBS's price thesis
- **Reputational risk:** ESG-driven capital reallocation increasingly targets opaque custody arrangements — Phase 6's 37% score is a systematic exposure in UBS's sustainable investing book

RETURN OPPORTUNITIES TRUEVALUE OPENS FOR UBS

- **Structural quality screen for Power & Resources:** TrueValue balance scores identify which commodity exposures within the theme will compound structurally vs which will face the extractive baseline's Years 7–10 fragility
- **Nature-linked product structuring:** Water and land credit markets represent $25–80M/year per major mine site — a product opportunity for UBS's structured products team aligned with the Decarbonization 5D
- **UHNW sustainable commodity products:** TrueValue-screened gold products can command a transparency premium with UBS's sustainability-focused UHNW client base — differentiated from standard gold exposure
- **First-mover advantage:** Supply chain transparency requirements will tighten under Swiss Ordinance, EU CSDDD, and equivalent regulation — UBS's early adoption of phase-level screening positions it ahead of the compliance curve
- **Africa/EM investment thesis:** The Shea chain's 91% balance score demonstrates that West African agricultural commodity investments structured with cooperative ownership outperform extractive equivalents — relevant to UBS's emerging markets allocation

THE THESIS IN ONE SENTENCE

UBS has already told its clients that gold is a buy at $5,000 and Power & Resources is a structural theme. The TrueValue Framework tells them which commodity supply chains within that theme will still be structurally sound in 2030 — and which will have compounded the hidden fragility that extractive, imbalanced supply chains always eventually reveal.

Ready to integrate TrueValue Framework analytics into UBS's CIO commodity research, sustainable investing screening, and Swiss Ordinance compliance processes? We provide phase-resolved balance analysis, structural quality scoring, supply chain transparency classification, and 10-year forward modelling for wealth management investment and compliance teams.

TrueValue Analytics
Structural Supply Chain Intelligence
Wealth Management & Sustainable Investing

APPENDIX

Methodology & Data Sources

DATA TRANSPARENCY NOTICE

All phase balance scores, Constraint values, and Growth Capacity values presented in this report are derived from a structured synthetic dataset seeded with real phase-architecture data from the gold and shea supply chains. Phase definitions, transparency classifications, structural relationships, and natural resource flow aspects are grounded in publicly available industry data, operational research, and national accounts. No specific entity-level financials are used. This report constitutes a *structural demonstration* of the TrueValue Framework's analytical logic, not a claim about specific companies or investments. UBS references are citations of published public research and do not imply any

FRAMEWORK CALCULATION

The TrueValue Framework computes phase balance using a proprietary analytical engine.

Balance Score = min(C, G) / max(C, G) × 100

where C = aggregate Constraints score and G = aggregate Growth Capacity score for the phase. Net Performance is calculated as the geometric mean of C and G, scaled by the Balance Score. Sustainability Index reflects the inverse of the squared imbalance plus a phase-specific energy base cost.

commercial relationship or UBS Group AG endorsement.

PHASE ARCHITECTURE SOURCES

- Gold supply chain phase definitions: LBMA Good Delivery standards; WGC supply chain transparency research; COMEX warehouse reporting
- Transparency classifications: structured transparency map derived from public disclosure analysis and industry research
- Shea supply chain data: Serious Shea / Acorn programme; Mirova feasibility research; Burkina Faso / Senegal value chain studies
- Triple Bottom Line benchmarks: Foran, Lenzen & Dey (2005), *Balancing Act: A Triple Bottom Line Analysis of the Australian Economy*, CSIRO / University of Sydney

UBS RESEARCH REFERENCES

- Commodities Outlook 2026: UBS Chief Investment Office
- Year Ahead 2026 "Escape Velocity?": UBS CIO, November 2025
- Gold Daily: "Gold should rally amid rising geopolitical tensions," UBS CIO, February 2026
- Wood Mackenzie mine production data (cited in UBS): 80 mines exhausting production plans by 2028
- Sustainability Report 2025: UBS Group AG
- Responsible Supply Chain Management Policy: UBS Group AG
- Human Rights Statement 2026: UBS Group AG
- Swiss Ordinance on Due Diligence and Transparency in Minerals and Metals from Conflict-Affected Areas — UBS compliance disclosure, 2025

PROJECTION ASSUMPTIONS

- Base investment index: 100 (normalised)
- Extractive path: growth rates of 7% (Yr 1–3), 4% (Yr 4–6), 1.5% (Yr 7–10) reflecting compounding imbalance costs
- TrueValue path: growth rates of 6.5% (Yr 1–3), 9% (Yr 4–6), 11.5% (Yr 7–10)
- Rates derived from structural efficiency model; not from historical financial time series
- Past structural patterns do not guarantee future performance

NOTE ON COMMODITY FRAMEWORK APPLICABILITY

The TrueValue Framework is commodity-agnostic. The gold supply chain is used as the primary case study given the depth of available phase-architecture data. The same framework applies to copper, aluminum, and agricultural commodities — all of which UBS's CIO identifies as facing structural supply shortage. A TrueValue analysis of any of these chains would produce a comparable phase-level risk and structural quality profile.

TrueValue Analytics — Structural Supply Chain Intelligence. This document is prepared for UBS's Chief Investment Office, Wealth Management, Sustainable Investing, and Compliance teams. It does not constitute an offer to buy or sell securities. Projections are illustrative structural models only.